

Exercise 1 - Kotlin Jetpack Compose Mobile Shop

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Course: COMP5450 Mobile Programming

Assignment Summary

This project implements a mobile shop app using Kotlin + Jetpack Compose. Users can browse products, open a detail page, choose color and size, add items to the bag, and view the shopping bag total. Product data is stored in memory (no database).

Implemented Features

- Landing/home screen inspired by the provided mockup.
- Category chips and product grid on the home screen.
- Product detail screen with rating, color selector, and size selector (EU/US/UK).
- Add to Bag flow from product detail and quick-add from grid.
- Bag screen listing selected items with remove action and total price calculation.

How to Run

Open exercises/exercise-1 in Android Studio.

Use Gradle JDK 17.

Wait for Gradle sync to complete.

Run the app on an Android emulator or physical device.

Optional CLI build: `export JAVA_HOME="$(brew --prefix openjdk@17)" && ./gradlew :app:assembleDebug`

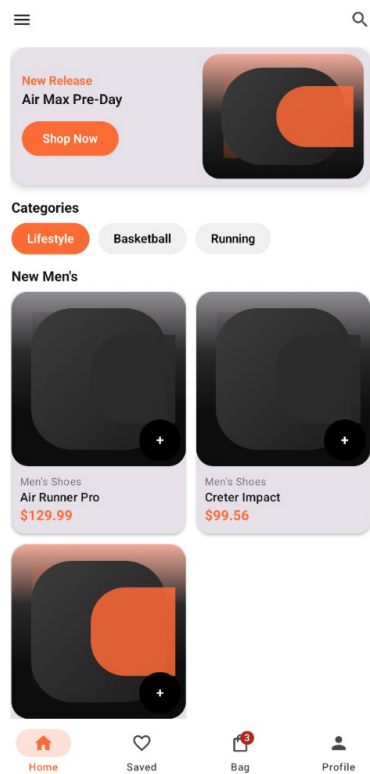
Project Structure (Key Files)

- MainActivity.kt - app entry point.
- data/Models.kt and data/SampleData.kt - product/cart models and sample data.
- ui/ShopViewModel.kt - state, filtering, cart logic, and total.
- ui/screens/* - home, detail, bag, and profile placeholder screens.

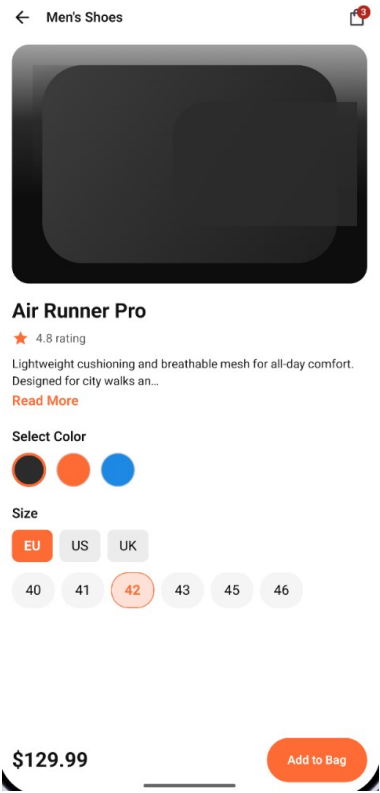
Runtime Screenshots

The following screenshots were captured from the running app on emulator/device.

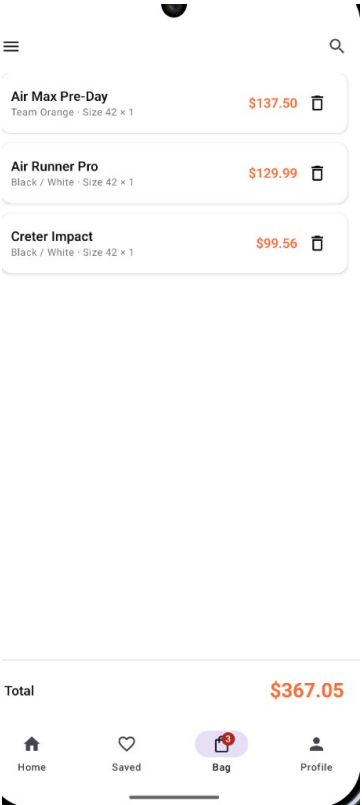
1) Home Screen



2) Product Detail Screen



3) Bag Screen with Total



4) Profile Screen (Placeholder)



Profile

Profile & settings (placeholder).

